

The Fourier Transform on Finite Abelian Groups

Math 419 Project

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Abstract

We will be looking at the development of Fourier Analysis on finite abelian groups. In parts (a) and (b), we deliver an exposition of the group theory and linear algebra needed for the development of this theory. Parts (c) and (d) setup the Hilbert spaces which the Fourier transform will be defined on. Part (e) explores the notion of group characters which play the role of trigonometric functions in this context. Finally, parts (f) and (g) define and explore the Fourier transform on finite abelian groups.

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Part A: Foundations of Group Theory

We begin in part A by establishing some foundational definitions and results from group theory that will be helpful in deriving results in later parts. We start by defining a *finite abelian group*

Definition 1 (Finite Abelian Group). A finite abelian group is a set G of finite cardinality, equipped with a binary operation $(\cdot) : G \times G \rightarrow G$, satisfying the following $\forall a, b, c \in G$:

- Associative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- Commutative: $a \cdot b = b \cdot a$
- Identity: $\exists e \in G, a \cdot e = a$
- Inverses: $\exists a^{-1} \in G, a \cdot a^{-1} = e$

We then continue by providing some examples of finite abelian groups.

Example 1. $\mathbb{Z}_5$, the set of integers modulo 5 under multiplication.

Example 2. The set $\{-1, 1\}$, under multiplication

Example 3. $\{0\}$, the zero group, under multiplication.

Definition 2 (Order of a Group). The *order* of a group G , denoted $|G|$, is the number of distinct elements in the group.

For example, the orders of the groups from example 1 - 3 are:

- $|\mathbb{Z}_5| = 5$.
- $|\{-1, 1\}| = 2$
- $|\{0\}| = 1$.

We now continue by giving some definitions that will play a role throughout the remaining parts.

Definition 3 (Group Homomorphism). A *group homomorphism* is a function ϕ between two groups $(G, \cdot)$ and $(H, *)$ such that $\forall a, b \in G, \phi(a \cdot b) = \phi(a) * \phi(b)$.

One important property of homomorphisms that arise from its operation preservation is that the identity in one group always maps to the identity in another group under a homomorphism. We provide this as a quick theorem.

Theorem 1. For any two groups G with identity element e_G and H with identity e_H and any homomorphism $\phi : G \rightarrow H, \phi(e_G) = e_H$.

Proof. Start by noting that $e_G \cdot e_G = e_G$ in G . Then:

$$\phi(e_G) = \phi(e_G \cdot e_G) = \phi(e_G)\phi(e_G)$$

This implies that:

$$\begin{aligned}\phi(e_G)[\phi(e_G)]^{-1} &= (\phi(e_G)\phi(e_G))[\phi(e_G)]^{-1} \\ e_H &= \phi(e_G)(\phi(e_G)[\phi(e_G)]^{-1}) \\ e_H &= \phi(e_G)e_H \\ e_H &= \phi(e_G)\end{aligned}$$

□

One very useful notion, defined below, is the notion of a *kernel* of a homomorphism.

Definition 4 (Kernel of a Homomorphism). For any groups G, H and any homomorphism $\phi : G \rightarrow H$, the set $\{x : \phi(x) = e, x \in G\}$ which contains all elements $x \in G$ such that $x \mapsto e$ where e is the identity element in H is called the *kernel* of ϕ and denoted $\ker(\phi)$.

We end this part with one more important definition.

Definition 5 (Group Isomorphism). Two groups G, H are said to be *isomorphic* if there exists a bijective homomorphism, called an *isomorphism*, $\phi : G \rightarrow H$ and we write $G \cong H$.

Part B: Foundations of Linear Algebra and $L^2(G)$

Let us begin this section with some foundation in linear algebra. These concepts will provide the tools needed to study the function space $L^2(G)$ and develop Fourier analysis on finite abelian groups.

Definition 6 ($\mathbb{C}$ -Vector Space). Let V be a set equipped with two operations: vector addition and scalar multiplication over $\mathbb{C}$. For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars $a, b \in \mathbb{C}$, the following axioms hold:

1. Associativity of addition: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
2. Commutativity of addition: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. Identity element of addition: There exists an element $\mathbf{0} \in V$, called the zero vector, such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$.
4. Inverse elements of addition: For every $\mathbf{v} \in V$, there exists an element $-\mathbf{v} \in V$, called the additive inverse of $\mathbf{v}$, such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
5. Compatibility of scalar multiplication with field multiplication: $a(b\mathbf{v}) = (ab)\mathbf{v}$
6. Identity element of scalar multiplication: $1\mathbf{v} = \mathbf{v}$, where 1 denotes the multiplicative identity in $\mathbb{C}$.
7. Distributivity of scalars: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
8. Distributivity of vectors: $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

Consider these examples of $\mathbb{C}$ -vector spaces:

- $\mathbb{C}^N$, the cartesian product of the complex numbers.
- $L^2(\mathbb{R})$, the set of square Lebesgue integrable functions on $\mathbb{R}$.
- $P_3(\mathbb{C})$, the set of third degree polynomials over the complex numbers.

Definition 7 (Basis of a Vector Space). A *basis* of a vector space V is a set of vectors $\{\mathbf{v}_i\} \subset V$, with the two following properties:

- Linear independence: $\sum_{i=1}^N c_i \mathbf{v}_i = \mathbf{0} \iff c_i = 0, \forall i = 1, 2, \dots, N$. That is to say no vector in the basis can be generated by a linear combination of the other vectors in the basis.
- $\forall \mathbf{u} \in V, \exists c_i \in \mathbb{C}$ such that $\mathbf{u} = \sum_i c_i \mathbf{v}_i$. That is, any vector in the vector space can be generated by a linear combination of the vectors in the basis.

Definition 8 (Dimension of a Vector Space). Given a vector space V and a basis set $\{\mathbf{v}_i\}_{1 \leq i \leq n}$, the *dimension* of a vector space is the size of $\{\mathbf{v}_i\}_{1 \leq i \leq n}$.

We now introduce the notions of dual spaces and homomorphisms, which will help us describe relationships between vector spaces.

Definition 9 (Dual of a Vector Space). The *dual* of a $\mathbb{C}$ -vector space V is the set V^* of all linear functions $\phi : V \rightarrow \mathbb{C}$.

Definition 10 (Homomorphism of Vector Spaces). A *homomorphism* of a vector space is a function ϕ between two vector spaces V and W on $\mathbb{C}$ such that $\forall a, b \in V, c \in \mathbb{C}, \phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ca) = c\phi(a)$.

Due to the operation preserving nature of homomorphisms, we see that, under a homomorphism, 0 is always mapped to 0 between vector spaces. This idea is presented in the following theorem.

Theorem 2. Given any two vector spaces V, W with $0_V, 0_W$ being 0 in the respective vector spaces, and a homomorphism $\phi : V \rightarrow W$, $\phi(0_V) = 0_W$.

Proof. Begin first by noting that $0_V + 0_V = 0_V$, then:

$$\begin{aligned}\phi(0_V + 0_V) &= \phi(0_V) \\ \phi(0_V) + \phi(0_V) &= \phi(0_V) \\ \phi(0_V) + \phi(0_V) - \phi(0_V) &= \phi(0_V) - \phi(0_V) \\ \phi(0_V) + 0_W &= 0_W \\ \phi(0_V) &= 0_W\end{aligned}$$

□

Definition 11 (Kernel of a Homomorphism). For any vector spaces V, W and any homomorphism $\phi : V \rightarrow W$, the set $\{x : \phi(x) = 0, x \in V\}$ which contains all elements $x \in V$ such that $x \mapsto 0$ where 0 is the identity element in W is called the *kernel* of ϕ and denoted $\ker(\phi)$.

Theorem 3. For any two vector spaces V with identity element 0_V and W with identity 0_W , then given any homomorphism $\phi : V \rightarrow W$, $\ker(\phi) = 0_V$ if and only if ϕ is injective.

Proof. We begin with the forward direction.

For any $a, b \in V$ with $\phi(a) = \phi(b)$ then:

$$\begin{aligned}\phi(a) &= \phi(b) \\ \implies \phi(a) - \phi(b) &= 0_W \\ \implies \phi(a - b) &= 0_W\end{aligned}$$

This implies $a - b \in \ker(\phi)$ but then clearly $a - b = 0$ so $a = b$, thus proving the forward direction. Next, for the backward direction, if $a \in \ker(\phi)$ then clearly $\phi(a) = 0_W$ but then $\phi(0_V) = 0_W$ hence $\phi(a) = \phi(0_V)$ and since ϕ is injective, $a = 0_V$. Thus proves the backward direction.

□

Definition 12 (Isomorphism of Vector Spaces). Two vector spaces V, W are said to be *isomorphic* if there exists a bijective homomorphism, called an *isomorphism*, $\phi : V \rightarrow W$ and we write $V \cong W$.

These fundamental concepts of linear algebra will now be applied to the function space $L^2(G)$, which will play a central role in our development of the Fourier transform.

Define $L^2(G) = \{f : G \rightarrow \mathbb{C}\}$, a finite dimensional vector space over $\mathbb{C}$. Then for every $a \in G$, let $\delta_a : G \rightarrow \mathbb{C}$ be defined by:

$$\delta_a(x) = \begin{cases} 0 & \text{if } x \neq a \\ 1 & \text{if } x = a \end{cases}$$

We now show that $\{\delta_a\}_{a \in G}$ forms a basis for $L^2(G)$.

Theorem 4. $\{\delta_a\}_{a \in G}$ forms a basis for $L^2(G)$

Proof. To establish linear independence, we look at two cases:

1. $|G| = 1$:

Trivially, $\{\delta_a\}_{a \in G}$ is linearly independent as $|\{\delta_a\}_{a \in G}| = 1$.

2. $|G| = N, N \geq 2$:

$\forall a_i \in G, 1 \leq i \leq N, i \in \mathbb{N}$ with $a_i \neq b_j$ if $i \neq j$, we have that the δ_{a_i} are linearly independent as, given $c_i \in \mathbb{C}$

$$\begin{aligned} & c_1 \delta_{a_1}(x) + c_2 \delta_{a_2}(x) + \cdots + c_k \delta_{a_k}(x) + \cdots + c_N \delta_{a_N}(x) = 0 \\ \implies & c_1 \delta_{a_1}(a_k) + c_2 \delta_{a_2}(a_k) + \cdots + c_k \delta_{a_k}(a_k) + \cdots + c_N \delta_{a_N}(a_k) = 0 \\ \implies & c_1 \cdot 0 + c_2 \cdot 0 + \cdots + c_k \cdot 1 + \cdots + c_N \cdot 0 = 0 \\ \implies & c_k = 0 \end{aligned}$$

Since k is arbitrary, we have that, evaluating at any $a_i \implies c_i = 0, \forall i \leq N$. Thus, $\{\delta_a\}_{a \in G}$ is linearly independent.

Now we show that $L^2(G)$ is spanned by $\{\delta_a\}_{a \in G}$. To do this, we show that $\forall f \in L^2(G)$:

$$f(x) = \sum_{a \in G} f(a) \delta_a(x)$$

We split this into two cases:

1. If $|G| = 1$ then clearly, since $x, a \in G \implies x = a$:

$$f(x) = f(a) \implies f(x) = f(a) \cdot 1 \implies f(x) = f(a) \delta_a(a) \implies f(x) = f(a) \delta_a(x)$$

2. If $|G| = N, N \geq 2$:

$\forall a_i \in G, 1 \leq i \leq N, i \in \mathbb{N}$ with $a_i \neq b_j$ if $i \neq j$ we have that, $\forall a_k \in G$:

$$\begin{aligned} & f(a_k) = f(a_k) \cdot 1 \\ \implies & f(a_k) = f(a_k) \delta_{a_k}(a_k) \\ \implies & f(a_k) = f(a_k) \delta_{a_k}(a_k) + 0 \\ \implies & f(a_k) = f(a_k) \delta_{a_k}(a_k) + [f(a_1) \delta_{a_1}(a_k) + \cdots + f(a_N) \delta_{a_N}(a_k)] \\ \implies & f(a_k) = f(a_1) \delta_{a_1}(a_k) + \cdots + f(a_k) \delta_{a_k}(a_k) + \cdots + f(a_N) \delta_{a_N}(a_k) \\ \implies & f(a_k) = \sum_{i=1}^N f(a_i) \delta_{a_i}(a_k) \\ \implies & f(a_k) = \sum_{a \in G} f(a) \delta_a(a_k) \\ \implies & f(x) = \sum_{a \in G} f(a) \delta_a(x), \text{ since } a_k \text{ is arbitrary} \end{aligned}$$

Thus we have shown that $\{\delta_a\}_{a \in G}$ is linearly independent and spans $L^2(G)$ and therefore forms a basis for $L^2(G)$. □

From this result, since $|\{\delta_a\}_{a \in G}| = |G|$ and since $\{\delta_a\}_{a \in G}$ forms a basis for $L^2(G)$, we also can conclude that $L^2(G)$ has dimension equal to $|G|$, presented in the corollary below.

Corollary 1. The dimension of $L^2(G)$ is $|G|$

Proof. Clearly, $|\{\delta_a\}_{a \in G}| = |G|$, so since this is a basis of $L^2(G)$, the dimension must be equal to $|G|$. □

From this, we also get the following result,

Corollary 2. The dimension of $L^2(\hat{G})$ is $|G|$

Proof. Since, by Corollary 1, the dimension of $L^2(\hat{G})$ is $|\hat{G}|$ and since, by Corollary 3, $|\hat{G}| = |G|$, the dimension of $L^2(\hat{G})$ is $|G|$. □

Part C: Integration

In this section, we define the notion of an *integral* over a set $U \subseteq G$ of a function $f \in L^2(G)$ and show some properties of this integral.

Definition 13 (Integral). The integral over a set $U \subseteq G$ of a function $f \in L^2(G)$, denoted $\int_U f$ is defined as:

$$\int_U f := \sum_{a \in U} f(a)$$

Example 4. To demonstrate this integral, we calculate $\int_U \delta_a$, for any $U \subseteq G$:

Suppose $|U| = M$ and $U = \{x_1, \dots, x_M\}$ then:

$$\begin{aligned} \int_U \delta_a &= \sum_{x \in U} \delta_a(x) \\ \implies \int_U \delta_a &= \delta_a(x_1) + \dots + \delta_a(x_M) \\ \implies \int_U \delta_a &= \begin{cases} 1 & \text{if } \exists x_i \in U \text{ s.t. } x_i = a \\ 0 & \text{otherwise} \end{cases} \\ \implies \int_U \delta_a &= \begin{cases} 1 & \text{if } a \in U \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Example 5. Another example is $\forall U \subseteq G, \forall a, b \in G$ to calculate $\int_U \delta_a \delta_b$:

$$\begin{aligned} \int_U \delta_a \delta_b &= \sum_{c \in U} \delta_a(c) \delta_b(c) \\ \implies \int_U \delta_a \delta_b &= 0 + \dots + \delta_a(a) \delta_b(a) + \dots + 0 + \dots + \delta_a(b) \delta_b(b) + \dots + 0 \\ \implies \int_U \delta_a \delta_b &= \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases} \end{aligned}$$

Two helpful properties that this integral satisfies is linearity.

Theorem 5. Integration over a subset U of a group G is linear, that is $\forall f_1, f_2 \in L^2(G), \forall c, d \in \mathbb{C}$:

$$\int_U cf_1 + df_2 = c \int_U f_1 + d \int_U f_2$$

Proof. The proof of Theorem 5 follows from the definition:

$$\begin{aligned} \int_U cf_1 + df_2 &= \sum_{a \in U} [cf_1(a) + df_2(a)] \\ \implies \int_U cf_1 + df_2 &= \sum_{a \in U} cf_1(a) + \sum_{a \in U} df_2(a) \\ \implies \int_U cf_1 + df_2 &= c \sum_{a \in U} f_1(a) + d \sum_{a \in U} f_2(a) \\ \implies \int_U cf_1 + df_2 &= c \int_U f_1 + d \int_U f_2 \end{aligned}$$

□

Another helpful property is presented in the following theorem.

Theorem 6. If U_1 and U_2 are two disjoint subsets of a group G then,

$$\int_{U_1 \cup U_2} f = \int_{U_1} f + \int_{U_2} f$$

Proof. Suppose $U_1 = \{a_1, \dots, a_n\}$ and $U_2 = \{b_1, \dots, b_m\}$ and are disjoint. Then,

$$\begin{aligned} \int_{U_1 \cup U_2} f &= \sum_{x \in U_1 \cup U_2} f(x) \\ \implies \int_{U_1 \cup U_2} f &= f(a_1) + \dots + f(a_n) + f(b_1) + \dots + f(b_m) \\ \implies \int_{U_1 \cup U_2} f &= \sum_{i=1}^n f(a_i) + \sum_{j=1}^m f(b_j) \\ \implies \int_{U_1 \cup U_2} f &= \sum_{a \in U_1} f(a) + \sum_{b \in U_2} f(b) \\ \implies \int_{U_1 \cup U_2} f &= \int_{U_1} f + \int_{U_2} f \end{aligned}$$

□

Part D: Inner Product

Using the notation of integration developed above, we now define an inner product and norm on $L^2(G)$. We start by defining the following mapping from $L^2(G)$ to $\mathbb{C}$.

$$\begin{aligned}\langle \cdot, \cdot \rangle : L^2(G) \times L^2(G) &\rightarrow \mathbb{C} \\ \langle f, g \rangle &\mapsto \int_G f \bar{g}\end{aligned}$$

Theorem 7. $\langle f, g \rangle$ defines an inner product in $L^2(G)$

Proof. We demonstrate that $\langle f, g \rangle$ satisfies all of the properties required of an inner product.

1. $\forall f, g \in L^2(G)$:

$$\begin{aligned}\langle f, g \rangle &= \int_G f \bar{g} = \sum_{a \in G} f(a) \overline{g(a)} \\ \implies \langle f, g \rangle &= \sum_{a \in G} \overline{g(a)} f(a) \implies \langle f, g \rangle = \int_G \bar{g} f \\ \implies \langle f, g \rangle &= \int_G \overline{\bar{g} f} \implies \langle f, g \rangle = \overline{\int_G g \bar{f}} \\ \implies \langle f, g \rangle &= \overline{\int_G g \bar{f}} \implies \langle f, g \rangle = \overline{\langle g, f \rangle}\end{aligned}$$

2. $\forall f, g, h \in L^2(G), \forall b, c \in \mathbb{C}$:

$$\begin{aligned}\langle bf + cg, h \rangle &= \int_G (bf + cg) \bar{h} = \int_G b f \bar{h} + c g \bar{h} \\ \implies \langle bf + cg, h \rangle &= b \int_G f \bar{h} + c \int_G g \bar{h} && \text{by Theorem 5} \\ \implies \langle bf + cg, h \rangle &= b \langle f, h \rangle + c \langle g, h \rangle\end{aligned}$$

3. $\forall f \in L^2(G), \exists c_a, d_a \in \mathbb{R}$ s.t. $\forall a \in G, f(a) = c_a + i d_a$ then,

$$\begin{aligned}\langle f, f \rangle &= \int_G f \bar{f} = \sum_{a \in G} f(a) \overline{f(a)} \\ \implies \langle f, f \rangle &= \sum_{a \in G} (c_a + d_a i)(c_a - d_a i) \\ \implies \langle f, f \rangle &= \sum_{a \in G} (c_a^2 + d_a^2) \geq 0\end{aligned}$$

As clearly, $c_a^2 + d_a^2 \geq 0, \forall c_a, d_a \in \mathbb{R}$

$$\implies \langle f, f \rangle \geq 0$$

4. $\forall f \in L^2(G), \exists c_a, d_a \in \mathbb{R} \text{ s.t. } \forall a \in G, f(a) = c_a + id_a \text{ and:}$

$$\begin{aligned}
& \langle f, f \rangle = 0 \\
& \iff \int_G f \bar{f} = 0 \\
& \iff \sum_{a \in G} f(a) \overline{f(a)} = 0 \\
& \iff \sum_{a \in G} (c_a^2 + d_a^2) = 0 \\
& \iff c_a = d_a = 0 \text{ as } c_a^2 + d_a^2 \geq 0, \forall c_a, d_a \in \mathbb{R} \\
& \iff f(a) = 0, \forall a \in G \\
& \iff f = 0
\end{aligned}$$

1-4 prove that, $\forall f, g \in L^2(G), \langle f, g \rangle$ satisfies the requirements to be an inner product. $\square$

Now that we have defined an inner product in $L^2(G)$ we see that the induced norm is given by:

$$\|f\|_{L^2(G)} = \sqrt{\int_G f \bar{f}} = \sqrt{\sum_{a \in G} f(a) \overline{f(a)}} \quad [1]$$

Since $L^2(G)$ is finite dimensional it is a complete inner-product vector space and thus a Hilbert space [1].

We next establish $\{\delta_a\}_{a \in G}$ as an orthonormal basis for $L^2(G)$:

Theorem 8. $\{\delta_a\}_{a \in G}$ forms an orthonormal basis for $L^2(G)$.

Proof. First, we start with orthogonality:

$\forall a, b \in G, a \neq b :$

$$\begin{aligned}
& \langle \delta_a, \delta_b \rangle = \int_G \delta_a \bar{\delta}_b \\
& \implies \langle \delta_a, \delta_b \rangle = \int_G \delta_a \delta_b \quad \text{since } \text{Im}(\delta_b) = 0 \implies \delta_b = \bar{\delta}_b \\
& \implies \langle \delta_a, \delta_b \rangle = 0 \quad \text{by Example 5}
\end{aligned}$$

Hence δ_a, δ_b are orthogonal $\forall a \neq b$

Second we show the, $\forall a \in G, \|\delta_a\|_{L^2(G)} = 1 :$

$$\begin{aligned}
& \langle \delta_a, \delta_a \rangle = \int_G \delta_a \bar{\delta}_a \\
& \implies \langle \delta_a, \delta_a \rangle = \int_G \delta_a \delta_a \quad \text{since } \text{Im}(\delta_a) = 0 \implies \delta_a = \bar{\delta}_a \\
& \implies \langle \delta_a, \delta_a \rangle = 1 \quad \text{by Example 5} \\
& \implies \|\delta_a\|_{L^2(G)} = \sqrt{\langle \delta_a, \delta_a \rangle} = \sqrt{1} = 1
\end{aligned}$$

Hence $\forall a \in G, \|\delta_a\|_{L^2(G)} = 1$

Therefore $\{\delta_a\}_{a \in G}$ forms an orthonormal basis for $L^2(G)$. $\square$

Part E: Characters and Orthogonality

We embark now in developing the theory of these characters for our purposes. Characters play a rôle in representation theory analogous to the rôle of trigonometric functions in classical Fourier analysis. We now present a helpful theorem concerning the dual of a group.

Definition 14 (Character of a Group). A *character* of a finite abelian group G is a homomorphism $\chi : G \rightarrow \mathbb{C}^\times$

Definition 15 (Dual of a Group). The *dual* of a finite abelian group G is the set of all characters of G , and is denoted $\hat{G}$

Theorem 9. $\hat{G}$ is a group.

Proof. We first show that the operation is associative. Consider any $\chi_1, \chi_2, \chi_3 \in \hat{G}$. We can make use of associativity of multiplication in $\mathbb{C}$.

$$\begin{aligned} ((\chi_1\chi_2)\chi_3)(a) &= (\chi_1\chi_2)(a)\chi_3(a) \\ &= \chi_1(a)\chi_2(a)\chi_3(a) \in \mathbb{C} \\ &= \chi_1(a)(\chi_2\chi_3)(a) \\ &= (\chi_1(\chi_2\chi_3))(a) \end{aligned}$$

Next we consider the character χ_0 where $\chi_0(a) = 1$ for all $a \in G$. We can show that this is the identity. Let χ_k be any character in $\hat{G}$.

$$(\chi_k\chi_0)(a) = \chi_k(a)\chi_0(a) = \chi_k(a) \cdot 1 = 1 \cdot \chi_k(a) = \chi_0(a)\chi_k(a) = \chi_0\chi_k(a)$$

Lastly we show that each character χ has an inverse χ^{-1} defined by $\chi^{-1}(a) = \chi(a^{-1})$, making use of χ being a homomorphism.

$$(\chi\chi^{-1})(a) = \chi(a)\chi^{-1}(a) = \chi(a)\chi(a^{-1}) = \chi(aa^{-1}) = \chi(1) = 1 = \chi_0(a)$$

□

Now that we have established that $\hat{G}$ is itself a group, we compare the structures of G and $\hat{G}$ in the following theorem.

Theorem 10. Let G be a finite abelian group. Then, $G \cong \hat{\hat{G}}$

Proof. See [2]

□

Corollary 3. If G is a finite abelian group then $|G| = |\hat{G}|$

Proof. From Theorem 10, we have that $G \cong \hat{\hat{G}}$, hence there exists bijections $\phi : G \rightarrow \hat{\hat{G}}$ and $\psi : \hat{\hat{G}} \rightarrow G$. Since ϕ and ψ must be onto their respective co-domains, we must have that $|G| = |\hat{\hat{G}}|$. □

Using the fact that a character is a map into $\mathbb{C}$, we can define the conjugate of a character as $\overline{\chi}(a) = \overline{\chi(a)}$. One can show that the conjugate of a character is equal to its inverse. For this, we need a better grasp of what the elements in the image of χ look like.

Lemma 1. Let $\chi \in \hat{G}$, then $|\chi(a)| = 1$, where G is a finite abelian group.

Proof. Suppose you form a list of powers of an element of G , that is consider $\{a, a^2, a^3, \dots\}$. Since groups must be closed under their own operation, and we only have finite elements, this infinite list must repeat itself. Thus for some powers $n, m, n \neq m$, it must be that $a^n = a^m$. Note that this means $n - m \neq 0$. So, we can consider $a^k = a^{n-m} = 1$. Call k the order of a .

Next consider that any element in $\mathbb{C}$ can be written as $re^{i\theta}$. So we can write $\chi(a) = re^{i\theta}$. Using the order equation from above, we get the following.

$$\begin{aligned} |\chi(a^k)| &= |\chi(1)| \\ |\chi(a)^k| &= |1| \\ |r^k e^{ik\theta}| &= 1 \\ |r^k| &= 1 \end{aligned}$$

We see that the final statement only holds for $r = \pm 1$ since $k \neq 0$. Thus $|\chi(a)| = |\pm e^{ik\theta}| = 1, \forall a \in G$. $\square$

Theorem 11. If χ is a character of a finite abelian group G , then $\bar{\chi} = \chi^{-1}$.

Proof. Recall that the norm of an element $z \in \mathbb{C}$ is given by $|z| = z\bar{z}$.

$$(\chi\bar{\chi})(a) = \chi(a)\bar{\chi}(a) = \chi(a)\overline{\chi(a)} = |\chi(a)|^2$$

And by Lemma 1, we know $|\chi(a)|^2 = 1 = \chi_0(a)$ for all $a \in G$. Thus $\bar{\chi} = \chi^{-1}$. $\square$

Following the establishment of Theorem 11, we prove two identities for the characters of G . We begin with a lemma.

Theorem 12. If G is a finite abelian group of order n with $a, b \in G$, $\chi, \chi_1, \chi_2 \in \hat{G}$, then the following hold:

$$\begin{aligned} \text{(a)} \quad \sum_{a \in G} \chi_1(a)\bar{\chi}_2(a) &= \begin{cases} 0 & \text{if } \chi_1 \neq \chi_2 \\ n & \text{if } \chi_1 = \chi_2 \end{cases} \\ \text{(b)} \quad \sum_{\chi \in \hat{G}} \chi(a)\bar{\chi}(b) &= \begin{cases} 0 & \text{if } a \neq b \\ n & \text{if } a = b \end{cases} \end{aligned}$$

Proof. We start with property (a) and split into two cases.

1. If $\chi_1 \neq \chi_2$ then we must have that $\chi_1\bar{\chi}_2 \neq \chi_0$ as if it were we would have that $\bar{\chi}_2 = \chi_1^{-1}$ and since, by Theorem 11, we have $\chi_1^{-1} = \bar{\chi}_1$ a clear contradiction of the hypothesis. Now, since $\chi_1\bar{\chi}_2 \neq \chi_0$, let $\chi = \chi_1\bar{\chi}_2$, $\exists c \in G$ such that $\chi(c) \neq 1$. Then,

$$\begin{aligned} \chi(c) \sum_{a \in G} \chi_1(a)\bar{\chi}_2(a) &= \chi(c) \sum_{a \in G} \chi(a) \\ &= \sum_{a \in G} \chi(a)\chi(c) \\ &= \sum_{a \in G} \chi(ac) \end{aligned}$$

Since $\forall b, c \in G$ we have that $\exists a \in G$ such that $b = ac$ and thus we have that, replacing a with ac only reorders the summation $\sum_{a \in G} \chi(a)$.

$$\begin{aligned} \implies \sum_{a \in G} \chi(ac) &= \sum_{a \in G} \chi(a) \\ \implies \sum_{a \in G} \chi(a)\chi(c) &= \sum_{a \in G} \chi(a) \\ \implies \left(\sum_{a \in G} \chi(a) \right) \chi(c) &= \sum_{a \in G} \chi(a) \end{aligned}$$

Since $\chi(c) \neq 1$:

$$\sum_{a \in G} \chi(a) = 0$$

2. If $\chi_1 = \chi_2$ then clearly:

$$\begin{aligned} \chi_1 \overline{\chi_2} &= \chi_1 \overline{\chi_1} \\ &= \chi_1 \chi_1^{-1} \quad \text{by Theorem 11} \\ &= \chi_0 \end{aligned}$$

Thus

$$\begin{aligned} \sum_{a \in G} \chi_1 \overline{\chi_1}(a) &= \sum_{a \in G} \chi_1 \chi_1^{-1}(a) \\ &= \sum_{a \in G} \chi_0(a) \\ &= \sum_{a \in G} 1 \\ &= n \end{aligned}$$

Now we prove property (b), proceeding by again splitting the proof into cases:

1. If $a \neq b$ then

$$\begin{aligned} \sum_{\chi \in \hat{G}} \chi(a) \overline{\chi}(b) &= \sum_{\chi \in \hat{G}} \chi(a) \chi^{-1}(b) \\ &= \sum_{\chi \in \hat{G}} \chi(a) \chi(b^{-1}) \\ &= \sum_{\chi \in \hat{G}} \chi(ab^{-1}) \end{aligned}$$

Now, let $ab^{-1} = c$. Then $\exists \chi^* \in \hat{G}$, $\chi^* \neq \chi_0$ such that $\chi^*(c) \neq 1$, and so we have:

$$\begin{aligned} \chi^*(c) \sum_{\chi \in \hat{G}} \chi(c) &= \sum_{\chi \in \hat{G}} \chi(c) \chi^*(c) \\ &= \sum_{\chi \in \hat{G}} (\chi \chi^*)(c) \end{aligned}$$

Then, $\forall \tilde{\chi} \in \hat{G}$, $\exists \chi$ such that $\chi \chi^* = \tilde{\chi}$, thus we have that replacing χ with $\chi \chi^*$ simply reorders the summation

$$\begin{aligned} \implies \sum_{\chi \in \hat{G}} (\chi \chi^*)(c) &= \sum_{\chi \in \hat{G}} \chi(c) \\ \implies \left(\sum_{\chi \in \hat{G}} \chi(c) \right) \chi^*(c) &= \sum_{\chi \in \hat{G}} \chi(c) \end{aligned}$$

Hence, since $\chi^*(c) \neq 1$ we must have:

$$\sum_{\chi \in \hat{G}} \chi(c) = 0$$

2. If $a = b$ then

$$\begin{aligned}\sum_{\chi \in \hat{G}} \chi(a) \overline{\chi}(a) &= \sum_{\chi \in \hat{G}} \chi(a) \chi^{-1}(a) \quad \text{by Theorem 11} \\ &= \sum_{\chi \in \hat{G}} \chi_0(a) \\ &= \sum_{\chi \in \hat{G}} 1 \\ &= |\hat{G}| \\ &= n\end{aligned}$$

By Corollary 3

□

Part F: Fourier Transform

Now, with the groundwork laid, we can begin defining the Fourier transform for finite abelian groups. First a few definitions, some analogous to those presented for groups.

We now present a lemma and a theorem concerning the dual of $L^2(G)$ for a finite abelian group G .

Lemma 2. For any abelian group G , the set $\{\delta_a^*\}_{a \in G}$, where:

$$\begin{aligned}\delta_a^* : L^2(G) &\rightarrow \mathbb{C} \\ f &\mapsto f(a)\end{aligned}$$

Defines a basis set for $(L^2(G))^*$.

Proof. We start by confirming that $\delta_a^* \in (L^2(G))^*, \forall a \in G$. Clearly, $\delta_a^* \in (L^2(G))^*$ as $\forall f, g \in L^2(G), a \in G, c, d \in \mathbb{C}$

$$\begin{aligned}\delta_a^*(cf + dg) &= (cf + dg)(a) \\ &= (cf)(a) + (dg)(a) \\ &= cf(a) + dg(a) \\ &= c\delta_a^*(f) + d\delta_a^*(g)\end{aligned}$$

Thus we see that δ_a^* is a linear function from $L^2(G)$ into $\mathbb{C}$. This forms a basis for $(L^2(G))^*$ as, let $n = |G|$, then:

(a) The set

$$\{\delta_{a_1}^*, \dots, \delta_{a_n}^*\}$$

is linearly independent. Consider the equation:

$$c_1\delta_{a_1}^* + \dots + c_n\delta_{a_n}^* = 0$$

where 0 is the function that maps every vector in $L^2(G)$ to 0. When evaluating this equation at any basis vector δ_{a_i} of $L^2(G)$ (Theorem 4), $1 \leq i \leq n$, we see:

$$\begin{aligned}(c_1\delta_{a_1}^* + \dots + c_i\delta_{a_i}^* + \dots + c_n\delta_{a_n}^*)(\delta_{a_i}) &= 0(\delta_{a_i}) \\ c_1\delta_{a_1}^*(\delta_{a_i}) + \dots + c_i\delta_{a_i}^*(\delta_{a_i}) + \dots + c_n\delta_{a_n}^*(\delta_{a_i}) &= 0 \quad \text{by the linearity of } \delta_a^* \\ c_1\delta_{a_i}(a_1) + \dots + c_i\delta_{a_i}(a_i) + \dots + c_n\delta_{a_i}(a_n) &= 0 \\ 0 + \dots + c_i + \dots + 0 &= 0 \\ c_i &= 0\end{aligned}$$

Since i is arbitrary and $\delta_{a_i}^*$ is linear, we see that, when evaluated at any vector $f(x) = \sum_{a \in G} f(a)\delta_a(x)$, this equation will be non-zero unless $c_1 = \dots = c_n = 0$. Thus $\{\delta_a^*\}_{a \in G}$ is linearly independent.

(b) $\forall \phi \in (L^2(G))^*$, define $d_i = \phi(\delta_{a_i})$ for $1 \leq i \leq n$. Next let

$$\psi := d_1\delta_{a_1}^* + \dots + d_n\delta_{a_n}^*$$

Now, $\forall f \in L^2(G)$, $f(x) = \sum_{b \in G} f(b)\delta_b(x)$, so:

$$\begin{aligned}
\psi(f) &= d_1\delta_{a_1}^*(f) + \cdots + d_n\delta_{a_n}^*(f) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= d_1\delta_{a_1}^*\left(\sum_{b \in G} f(b)\delta_b(x)\right) + \cdots + d_n\delta_{a_n}^*\left(\sum_{b \in G} f(b)\delta_b(x)\right) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= d_1 \sum_{b \in G} \delta_{a_1}(f(b)\delta_b(x)) + \cdots + d_n \sum_{b \in G} \delta_{a_n}(f(b)\delta_b(x)) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= d_1 \sum_{b \in G} f(b)\delta_{a_1}(\delta_b(x)) + \cdots + d_n \sum_{b \in G} f(b)\delta_{a_n}(\delta_b(x)) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= d_1 \sum_{b \in G} f(b)(\delta_b(a_1)) + \cdots + d_n \sum_{b \in G} f(b)(\delta_b(a_n)) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= d_1 f(a_1) + \cdots + d_n f(a_n)
\end{aligned}$$

Then turning our d_i back to the $\phi(\delta_{a_i})$:

$$\begin{aligned}
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= \phi(\delta_{a_1}(x))f(a_1) + \cdots + \phi(\delta_{a_n}(x))f(a_n) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= \phi(f(a_1)\delta_{a_1}(x)) + \cdots + \phi(f(a_n)\delta_{a_n}(x)) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= \phi(f(a_1)\delta_{a_1}(x) + \cdots + f(a_n)\delta_{a_n}(x)) \\
\psi\left(\sum_{b \in G} f(b)\delta_b(x)\right) &= \phi\left(\sum_{b \in G} f(b)\delta_b(x)\right) \quad =
\end{aligned}$$

Thus, since ϕ is arbitrary, we see that the set $\{\delta_a^*\}_{a \in G}$ spans $(L^2(G))^*$.

We see now, by (a) and (b) that $\{\delta_a^*\}_{a \in G}$ forms a basis for $(L^2(G))^*$. □

Theorem 13. For any abelian group G , $L^2(G) \cong (L^2(G))^*$.

Proof. Define the following mapping on the bases of $L^2(G)$ and its dual:

$$\begin{aligned}
\Phi : L^2(G) &\rightarrow (L^2(G))^* \\
\delta_a &\mapsto \delta_a^*
\end{aligned}$$

Then extend this to map to any $f \in L^2(G)$:

$$\Phi(f) = \sum_{a \in G} f(a)\delta_a^*$$

Now, we show that this forms an isomorphism.

We begin by showing that this is, in fact, a homomorphism.

(a) $\forall f, g \in L^2(G)$,

$$\begin{aligned}
\Phi(f + g) &= \sum_{a \in G} (f(a) + g(a))\delta_a^* \\
&= \sum_{a \in G} f(a)\delta_a^* + \sum_{a \in G} g(a)\delta_a^* \\
&= \sum_{a \in G} f(a)\delta_a^* + \sum_{a \in G} g(a)\delta_a^* \\
&= \Phi(f) + \Phi(g)
\end{aligned}$$

(b) $\forall f \in L^2(G), c \in \mathbb{C}$,

$$\Phi(cf) = \sum_{a \in G} cf(a)\delta_a^* = c \sum_{a \in G} f(a)\delta_a^* = c\Phi(f)$$

(a) and (b) show that this is a homomorphism.

To show that this is bijective, we show that it is both surjective and injective.

(a) To show surjectivity, since the basis of $L^2(G)$ is $\{\delta_a\}_{a \in G}$ and the basis of $(L^2(G))^*$ is $\{\delta_a^*\}_{a \in G}$ and since Φ maps each $\delta_a \mapsto \delta_a^*$, and since Φ is a homomorphism, any linear combination from $\{\delta_a^*\}_{a \in G}$ can be mapped to by a linear combination from $\{\delta_a\}_{a \in G}$ with the same coefficients, hence Φ is onto.

(b) To show injectivity, we appeal to Theorem 3. Suppose there is some $f \in L^2(G)$ such that $f \in \ker(\Phi)$. Then, since $f = \sum_{a \in G} f(a)\delta_a$,

$$\Phi(f) = \Phi\left(\sum_{a \in G} f(a)\delta_a\right) = \sum_{a \in G} \Phi(f(a)\delta_a) = \sum_{a \in G} f(a)\Phi(\delta_a) = \sum_{a \in G} f(a)\delta_a^*$$

Thus we have that, given $f \in \ker(\Phi)$,

$$\sum_{a \in G} f(a)\delta_a^* = 0$$

Since $\{\delta_a^*\}_{a \in G}$ is a basis for $(L^2(G))^*$, we must have that $f(a) = 0$ for all $a \in G$ and so $f = 0$. As such, $\ker(\Phi) = \{0\}$

As such, (a) and (b) prove that Φ is a bijection, thus Φ is an isomorphism and so, since we have demonstrated an isomorphism between $L^2(G)$ and $(L^2(G))^*$, we arrive at the result, $L^2(G) \cong (L^2(G))^*$. $\square$

Lemma 3. The set $\hat{G}$ forms a basis for $L^2(G)$.

Proof. Clearly, $\forall \chi \in \hat{G}$, we have $\chi \in L^2(G)$ as, by definition $\chi : G \rightarrow \mathbb{C}$. Clearly, the set $\hat{G}$ is linearly independent, as, let $n = |G| = |\hat{G}|$, for $c_j \in \mathbb{C}, 1 \leq j \leq n$:

$$\begin{aligned} & \sum_{j=1}^n c_j \chi_j = 0 \\ \implies & \left\langle \sum_{j=1}^n c_j \chi_j, \chi_k \right\rangle = \langle 0, \chi_k \rangle \\ \implies & \sum_{j=1}^n c_j \langle \chi_j, \chi_k \rangle = 0 && \text{by Theorem 7} \\ \implies & \sum_{j=1}^n c_j \left(\sum_{a \in G} \chi_j \overline{\chi_k} \right) = 0 \\ \implies & \sum_{j=1}^n c_j \cdot n \cdot \delta_{j,k} = 0 && \text{by Theorem 12} \\ \implies & nc_k = 0 \\ \implies & c_k = 0 \end{aligned}$$

Since k is arbitrary, clearly $c_j = 0, \forall j$, hence linear independence. Then, to show $\hat{G}$ spans $L^2(G)$, for any

$f \in L^2(G)$, since $\langle f, \chi \rangle \in \mathbb{C}$:

$$\begin{aligned}
\sum_{\chi \in \hat{G}} \frac{1}{|G|} \langle f, \chi \rangle \chi &= \frac{1}{|G|} \sum_{\chi \in \hat{G}} \left(\sum_{a \in G} f(a) \overline{\chi(a)} \right) \chi(x) \\
&= \frac{1}{|G|} \sum_{a \in G} f(a) \left(\sum_{\chi \in \hat{G}} \chi(x) \overline{\chi(a)} \right) \\
&= \frac{1}{|G|} \sum_{a \in G} f(a) \cdot |G| \cdot \delta_{a,x} && \text{by Theorem 12 (b)} \\
&= \frac{1}{|G|} |G| f(x) \\
&= f
\end{aligned}$$

Thus, $\hat{G}$ forms a linearly independent spanning set and therefore forms a basis for $L^2(G)$. $\square$

With the characters of G now established as a basis for $L^2(G)$, we can use this structure to further develop the relationship between $L^2(G)$ and $L^2(\hat{G})$.

Theorem 14. The dual of $L^2(G)$ is $L^2(\hat{G})$.

Proof. Start by defining the function:

$$\begin{aligned}
\Phi : L^2(G) &\rightarrow L^2(\hat{G}) \\
f &\mapsto \hat{f}
\end{aligned}$$

Where,

$$\hat{f}(\chi) = \langle f, \chi \rangle$$

We demonstrate that this is a homomorphism. For any $\chi, f, g \in L^2(G)$ and $c, d \in \mathbb{C}$:

$$\begin{aligned}
\Phi(cf + dg) &= \widehat{(cf + dg)}(\chi) \\
&= \langle cf + dg, \chi \rangle \\
&= c\langle f, \chi \rangle + d\langle g, \chi \rangle \\
&= c\hat{f}(\chi) + d\hat{g}(\chi) \\
&= c\Phi(f) + d\Phi(g)
\end{aligned}$$

Then, we can see that this is a bijection as:

(a) For any f such that $\Phi(f) = 0$, we see:

$$0 = \Phi(f) = \langle f, \chi \rangle$$

From Theorem 3, we see that:

$$f(x) = \sum_{\chi \in \hat{G}} \langle f, \chi \rangle \chi = \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi = \sum_{\chi \in \hat{G}} \Phi(f) \chi$$

Thus, if $\Phi(f) = 0$, we have:

$$f(x) = \sum_{\chi \in \hat{G}} \Phi(f) \chi = \sum_{\chi \in \hat{G}} 0 \cdot \chi = 0$$

Hence, $\ker(\Phi) = \{0\}$ and so Φ is injective.

(b) Since Φ is injective any element in $L^2(G)$ must get mapped to a unique element in $L^2(\hat{G})$, and since the dimensions of $L^2(G)$ and $L^2(\hat{G})$ are the same, Φ must be a surjection.

Thus (a) and (b) prove that Φ is a bijection, and so Φ is an isomorphism, and we may conclude that $L^2(G) \cong L^2(\hat{G})$. Combining with Theorem 13, we get that $(L^2(G))^* \cong L^2(G) \cong L^2(\hat{G})$, which shows that the dual of $L^2(G)$ is $L^2(\hat{G})$. $\square$

Motivated by this relation between $L^2(G)$ and $L^2(\hat{G})$, we then take the isomorphism from the proof of Theorem 14 and define the Fourier Transform as follows,

Definition 16 (Fourier Transform). The *Fourier Transform* of a function $f \in L^2(G)$ is defined pointwise by:

$$\begin{aligned}\mathcal{F} : L^2(G) &\rightarrow L^2(\hat{G}) \\ f &\mapsto \hat{f}(\chi) = \langle f, \chi \rangle\end{aligned}$$

Furthermore, we take inspiration from Theorem 3 and the proof of Theorem 14 in defining the Fourier Inversion formula below.

Definition 17 (Fourier Inversion Formula). The *Fourier Inversion* of a function $\hat{f} \in L^2(\hat{G})$ is defined by:

$$\begin{aligned}\mathcal{F}^{-1} : L^2(\hat{G}) &\rightarrow L^2(G) \\ \mathcal{F}^{-1}(\hat{f}) &= \frac{1}{|\hat{G}|} \sum_{\chi \in \hat{G}} \hat{f}(\chi) \chi\end{aligned}$$

Corollary 4. The Fourier Inversion formula is the inverse of the Fourier Transform.

Proof. Let $|G| = |\hat{G}| = n$. For any $\chi, f \in L^2(G)$, taking the composition $\mathcal{F} \circ \mathcal{F}^{-1}$ we see:

$$\begin{aligned}\mathcal{F}^{-1}(\mathcal{F}(f(x))) &= \mathcal{F}^{-1}(\langle f, \chi \rangle) \\ &= \frac{1}{n} \sum_{\chi \in \hat{G}} \langle f, \chi \rangle \chi(x) \\ &= \frac{1}{n} \sum_{\chi \in \hat{G}} \left(\sum_{a \in G} f(a) \overline{\chi(a)} \right) \chi(x) \\ &= \frac{1}{n} \sum_{a \in G} f(a) \left(\sum_{\chi \in \hat{G}} \overline{\chi(a)} \chi(x) \right) \\ &= \frac{1}{n} \sum_{a \in G} f(a) \cdot n \cdot \delta_{a,x} && \text{by Theorem 12 (b)} \\ &= \sum_{a \in G} f(a) \delta_{a,x} \\ &= f(x)\end{aligned}$$

$\square$

Theorem 15 (Plancherel's Identity). Let $f \in L^2(G)$, then the following holds:

$$\|f\|_{L^2(G)} = \frac{1}{\sqrt{n}} \|\hat{f}\|_{L^2(\hat{G})}$$

Proof.

$$\begin{aligned}
\|\hat{f}\|_{L^2(\hat{G})}^2 &= \sum_{\chi \in \hat{G}} \hat{f}(\chi) \overline{\hat{f}(\chi)} = \sum_{\chi \in \hat{G}} \left(\sum_{a \in G} f(a) \overline{\chi(a)} \right) \overline{\left(\sum_{b \in G} f(b) \overline{\chi(b)} \right)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{a \in G} f(a) \overline{\chi(a)} \right) \left(\sum_{b \in G} \overline{f(b)} \chi(b) \right) \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{a \in G} \sum_{b \in G} \overline{f(b)} \chi(b) f(a) \overline{\chi(a)} \right) \\
&= \sum_{a \in G} \sum_{b \in G} \overline{f(b)} f(a) \sum_{\chi \in \hat{G}} \chi(b) \overline{\chi(a)}
\end{aligned}$$

At this point, we may notice the applicability of Theorem 12, where the entire term in the triple summation results in a 0 if $a \neq b$, leaving only those terms where $a = b$, which will have a factor of n .

$$\begin{aligned}
\|\hat{f}\|_{L^2(\hat{G})}^2 &= \sum_{a \in G} \overline{f(a)} f(a) n = n \|f\|_{L^2(G)}^2 \\
\implies \|f\|_{L^2(G)} &= \frac{1}{\sqrt{n}} \|\hat{f}\|_{L^2(\hat{G})}
\end{aligned}$$

□

Part G: Convolution

We now define the translation of a function in $L^2(G)$ and examine its effect on the Fourier transform.

Definition 18 (Translation). Let $a \in G$ and $f \in L^2(G)$. The translation of f by a is the function $\tau_a f \in L^2(G)$ defined by

$$\tau_a f(x) = f(xa^{-1}),$$

where we are considering a multiplicative group.

We now present a theorem concerning the Fourier transform of a translated function.

Theorem 16. Let $a \in G$, $f \in L^2(G)$, and $\chi \in \widehat{G}$. Then

$$\widehat{\tau_a f}(\chi) = \overline{\chi(a)} \widehat{f}(\chi).$$

Proof. By the definitions of the Fourier transform and translation,

$$\begin{aligned} \widehat{\tau_a f}(\chi) &= \sum_{x \in G} \tau_a f(x) \overline{\chi(x)} \\ &= \sum_{x \in G} f(xa^{-1}) \overline{\chi(x)}. \end{aligned}$$

Let $y = xa^{-1}$, so that $x = ya$. Since the map $x \mapsto xa^{-1}$ is a bijection on G , replacing x by y only reorders the summation. Thus,

$$\begin{aligned} \widehat{\tau_a f}(\chi) &= \sum_{y \in G} f(y) \overline{\chi(ya)} \\ &= \sum_{y \in G} f(y) \overline{\chi(y)} \overline{\chi(a)} \\ &= \overline{\chi(a)} \sum_{y \in G} f(y) \overline{\chi(y)} \\ &= \overline{\chi(a)} \widehat{f}(\chi). \end{aligned}$$

□

We now turn from translation to another important operation on functions in $L^2(G)$, known as convolution.

Definition 19 (Convolution). For $f, g \in L^2(G)$, define their convolution by

$$(f * g)(x) = \sum_{y \in G} f(y) g(xy^{-1}).$$

We now show that convolution is commutative.

Theorem 17. For every $f, g \in L^2(G)$,

$$f * g = g * f.$$

Proof. For $x \in G$,

$$(f * g)(x) = \sum_{y \in G} f(y) g(xy^{-1}).$$

Let $z = xy^{-1}$. Then

$$y = z^{-1}x.$$

Since G is abelian,

$$z^{-1}x = xz^{-1}.$$

As y ranges over G , so does z . Therefore,

$$\begin{aligned} (f * g)(x) &= \sum_{z \in G} f(xz^{-1})g(z) \\ &= \sum_{z \in G} g(z)f(xz^{-1}) \\ &= (g * f)(x). \end{aligned}$$

Hence,

$$f * g = g * f.$$

□

Finally, we show that the Fourier transform converts convolution into pointwise multiplication.

Theorem 18. For every $f, g \in L^2(G)$ and $\chi \in \widehat{G}$,

$$\widehat{f * g}(\chi) = \widehat{f}(\chi)\widehat{g}(\chi).$$

Proof. For $\chi \in \widehat{G}$,

$$\begin{aligned} \widehat{f * g}(\chi) &= \sum_{x \in G} (f * g)(x) \overline{\chi(x)} \\ &= \sum_{x \in G} \sum_{y \in G} f(y)g(xy^{-1}) \overline{\chi(x)}. \end{aligned}$$

Let $z = xy^{-1}$, so that $x = zy$. Then

$$\begin{aligned} \widehat{f * g}(\chi) &= \sum_{y \in G} \sum_{z \in G} f(y)g(z) \overline{\chi(zy)} \\ &= \sum_{y \in G} \sum_{z \in G} f(y)g(z) \overline{\chi(z)} \overline{\chi(y)} \\ &= \left(\sum_{y \in G} f(y) \overline{\chi(y)} \right) \left(\sum_{z \in G} g(z) \overline{\chi(z)} \right) \\ &= \widehat{f}(\chi) \widehat{g}(\chi). \end{aligned}$$

Therefore,

$$\widehat{f * g}(\chi) = \widehat{f}(\chi)\widehat{g}(\chi).$$

□

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